

A Historical Analysis on Interoperability in Estonian Data Exchange Architecture: Perspectives from the Past and for the Future

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ABSTRACT

The importance of interoperability for enabling e-governance and e-service provision cannot be overstated. In Estonia, the interoperability data exchange platform, X-Road, has been implemented since 2001 and was integrated with the Finnish public sector in 2018. In the context of the EU, it is an exceptional case of a cross-border interoperability platform. However, the Estonian government has proposed next generation government e-services using virtual assistants. This may necessitate potential changes to X-Road if the members from its collectively governed body, Nordic Institute for Interoperable Services (NIIS) decide to do so. Thus, this paper seeks to understand how X-Road developed historically related to European Interoperability Framework principles while also providing an outlook for the future based upon this next generation e-service concept. An exploratory case-study approach was adopted, and six semi-structured interviews were conducted with the architects, developers and public sector officials of the original and current version of X-Road. A thematic analysis was then applied. Based on this analysis, the decision to decentralize the initial X-Road helped create sustainable interoperability supported by legislation written at the time and since then. Future work is presented involving the integration of microservices on an abstract level.

CCS CONCEPTS

• **information systems**; • **applied computing** → enterprise computing; enterprise interoperability; information integration and interoperability;

KEYWORDS

e-governance, interoperability, data exchange, x-road

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1 INTRODUCTION

Interoperability is more than just technical specifications — it is a cornerstone of e-government, encompassing legal frameworks, semantics, ontologies, organizations and trust, among other variables [1]. Subsequently, the ability of governmental organizations to exchange data internally and externally unlocks the potential for creating digital services that are open, efficient, efficacious and citizen-centric [2]. When interoperability is not ensured, there may be serious consequences. In nationalized healthcare infrastructure, a lack of standardization and interoperable Electronic Health Records has led to medical errors, diminishing the well-being of patients [3].

In the EU nation of Estonia, the X-Road has been an established interoperable data exchange framework since 2001. It is a distributed, unified data exchange layer connecting over 600 Estonian private and public sector organizations over the public internet. X-Road handles nearly one billion queries annually [4] and operates under an MIT open-source license. For comparison, the EU's interoperable document exchange platform, eDelivery, which has exchanged almost half a billion documents since 2015 [5].

Although X-Road was originally coordinated by the state information system department (RISO) of the Ministry of Economic Affairs and later by the Estonian State Information System Authority (RIA), since 2017 the non-profit Nordic Institute for Interoperable Solutions (NIIS) is the governing entity of the X-Road core software (the national instances are governed by the authorities of each country). As a built-in feature, X-Road since version 6 supports cross-border data exchange through federative agreements. In 2018, this was actualized when the Estonian and Finnish X-Road ecosystems were connected to one another [6]. While the exact scope of value creation from this exchange is limited, valid interoperable cross-border exchanges in the EU and consequently, the connected X-Road ecosystems provide a real-life example of data interoperability between nation-states. X-Road instances have also been deployed globally in Iceland, Columbia, Argentina, Japan, the Faroe Islands and in other countries respectively [4].

Thus, it is fair to describe X-Road as a successful enabler of data exchange use-cases in Estonia and abroad.

However, although most services in Estonia are paperless, the Government Chief Information Office (GCIO) recognizes the provision of next generation e-services necessitates structural upgrades to X-Road. In 2020 the GCIO of Estonia published a white paper entitled “Next Generation Digital Government Architecture” outlining a strategic vision for implementing proactive, artificially intelligent e-services, termed the KrattAI initiative [7]. The cornerstone of this vision is developing virtual agents for automation of domestic and cross-border public services [7]. For example, a virtual assistant

would proactively inform a citizen that their passport is expiring and automatically place an order for a new one. Incorporating AI into decision-making like in the scenario above is challenging [8], but to empower scalable, proactive virtual-agent-driven cross-border e-services as KrattAI intends, certain architectural changes to X-Road warrant investigation.

While X-Road has been explored in academic literature, it has primarily been conducted through comparative analysis [9] or specific aspects of the exchange layer have been investigated like private-public partnerships [10] and digital signature schema [11]. In contrast, the novelty of this research is presenting an analysis of X-Road from a historical perspective, while also investigating X-Road's compatibility with next generation digital architectures like KrattAI and microservices.

This paper has two objectives. First is analyzing the context at the end of the 1990s and the beginning of the early 2000s which enabled initial X-Road implementation and adoption based on interoperability principles. Through analyzing the premises of X-Road's initial development from interviewing Estonian policymakers, private sector actors and requirements architects who were there in the beginning, we better understand X-Road's evolution and potential scalability for future digital technologies. Second, this research seeks to understand X-Road's current interoperable capacity for the proposed Estonian next generation digital government architecture through interviewing the current CTO of NIIS. Based on these objectives, two research questions are proposed:

RQ1: How did the historical conditions related to interoperability principles affect X-Road implementation and adoption in Estonia?

RQ2: How is the current X-Road ecosystem equipped for the future of next generation government services?

In order to analyze this phenomenon, this research first presents the European Interoperability Framework (EIF) in the theoretical overview. Next, semi-structured interview methods and thematic coding based on the EIF is justified. Then the proceeding section presents the results of the interviews and discusses implications. Lastly, future work is provided to conclude the paper.

2 THEORETICAL OVERVIEW

The definition of interoperability has been proposed by many researchers and institutions in the fields of IT, public administration and healthcare, etc. The IEEE provides a simplistic definition of interoperability: "the ability of two or more systems or components to exchange information and to use the information that has been exchanged" [12]. Interoperability is also well summarized by the European Interoperability Framework: "the ability of organizations to interact towards mutually beneficial goals, involving the sharing of information and knowledge between these organizations, through the business processes they support, by means of the exchange of data between their ICT systems". The EIF has further deconstructed interoperability into four levels: technical, semantic, organizational and legal [13].

Technical interoperability is a primary driver for providing e-services and from a general perspective "covers the applications and infrastructures linking systems and services" [14]. More specifically, it dictates the ability for subsystems to interface with one another through standardized practices, shared technical standards

and frameworks [15]. The total aggregate of subsystem interconnectivity is what provides tremendous business value for the public sector. Data is also a critical component to technical interoperability. It's transformation from one standard to another, the security of its transport between different entities through Hypertext Transfer Protocol Secure (HTTPS) and various data integration services fuel decision-making in the public and private sector [16].

Further related to data is semantic interoperability. This level pertains to both the "semantic and syntactic aspects of data" [14]. On a basic level, semantics relate to everyone in a data exchange ecosystem speaking the same language; where data definitions and the corpus of words used to describe processes and information are all understandable [14]. Data has to be precise and error-free when exchanged, abiding by the correct syntax demanded by a compiler. Consequently, the deployment of different types of syntactical analyzers mitigates the potential for errors to occur [17]. While syntax has a specific and important function, semantic standardization was chosen for interoperability analysis in this paper.

The concept of organizational interoperability is closely connected to the semantic domain of data. Cooperative relationships between organizations are influenced by common business goals and agreeing to semantic standardization [14]. Internally, regulations and policies guide organizational adoption of interoperability with other organizations. Users also have representation in the organizational interoperability paradigm. The ability for organizations to provide effective and efficient services that meet the needs of users is also an important component to interoperability [14].

In society with the rule of law, organizations operate under legislation established by the public sector or sometimes in the case of the EU, supranational level. This impacts interoperability in a variety of ways. Data protection acts, legislation on digital signature requirements and its acceptability, e-identity and other legal frameworks shape the way interoperability occurs. In sum, the four levels of the EIF are the foundation for "integrated public service governance" [13]. In Estonia, X-Road ensures integrated public e-services like e-invoicing [18] occur at the local [19] and national level. As the IEF definition embodies a holistic approach to interoperability, it is the primary thematic coding framework applied in this research.

3 METHODOLOGY

In this paper, the authors deploy qualitative methods via an exploratory case-study approach. Case studies can embody qualitative methods by exploring specific phenomena through the triangulation of different data sources: interviews, focus groups, external articles, etc. [20]. A critical component to this research was the deployment of qualitative, semi-structured interviews with relevant stakeholders. Semi-structured interviews provide researchers a framework to talk with people in a 'self-conscious, orderly and partially structured' way [21]. The open-ended format of this interviewing technique allows for flexibility and the ability for interviewees to reflect and express themselves to the fullest intent possible. In this research, we conducted six semi-structured remote interviews averaging roughly an hour each. Figure 1 describes the interviewee affiliations and roles related to X-Road's development in greater detail.

Interviewee Number	X-Road Role	Stakeholder Sector
Interviewee 1	e-Government expert and consulted on X-Road version one in the public sector	Former CIO of Estonia
Interviewee 2	Primary Architect of the X-Road version one	Private Sector
Interviewee 3	Head Co-Manager of X-Road version one	Private Sector
Interviewee 4	Interoperability expert who works on current and previous versions of X-Road	Private Sector
Interviewee 5	Technical advisor for X-Road version one	Private Sector
Interviewee 6	Currently the CTO of NIIS	NIIS

To validate our interview questions, we approached Interviewee 1 who an Estonian expert on e-governance is and was an influential proponent of X-Road in the Estonian public sector. After we identified the founders of X-Road from technical, legal, and organizational perspectives through research and networking, we contacted the interviewees and asked fifteen different questions about the historical conditions at the time related to the EIF interoperability principles. Additionally, we asked about X-Road’s current and future outlook with next generation government e-services.

In order to gain greater qualitative insight into the X-Road phenomena, analytical thematic analysis was manually applied to the semi-structured interview transcripts through Otter.ai transcription service. There are two overarching approaches for conducting thematic analysis: concept-driven and data-driven. Concept driven takes ideas from previous theoretical literature and contextual information to form a framework for analysis [22]. Therefore, a list of keywords should be constructed beforehand and described before coding is conducted. On the other hand, data-driven coding doesn’t start with a framework and instead starts with zero codes, letting the interviews organically form applicable topics [23].

In this paper, a concept-driven coding approach was adopted for answering the first research question. The authors use the four interoperability principles established by the EIF and described in the theoretical overview. For answering the second research question, a data-driven approach was taken. Analytical coding was applied to the transcripts. Gibbs [23] Describes analytical coding as containing more complex labeling of because it is based on the interpretation of the interviewer [23]. In sum, this research uses an exploratory case study approach by triangulating six interview transcripts, external articles and thematic analysis.

4 RESULTS

Historically, the Estonian public sector in the late 90s and early 2000s was transitioning from Sovietization to a newly re-independent country. As Interviewee 1 described, Estonia’s public sector digital capacity during this period amounted to a few IBM components and a lack of IT human capital. Interviewee 4 further expanded on the public sector situation at the time: “real time, dynamic cooperation between agencies, was a real bottleneck. And one reason definitely was that some of the registries, most notably our population registry was building some kind of business model around. I would say selling or providing paid access to this data.”

Hence, it was recognized this was an unsustainable practice for the development of e-governance in Estonia. According to Interviewee 3, in 1998-99 there were various public-private sector consultations trying to improve the situation: “many ministries found that they need to integrate their IT systems, and each of these ministries asked from the government to finance this. So, if we have ten different ministries, then we need to do ten integrations and fund all of them”. However, the state did not have the funds at the time, and pragmatically there were “ideas to do something centrally and interconnect these ministries with one middleware.”

4.1 Technical Interoperability

This led to two competing perspectives related to the architecture of X-Road’s technical interoperability. On the one hand, the Estonian government’s procurement for such a system called for a highly centralized system using Service Bus architecture which makes data format conversions easier. Yet there was a consortium of private sector companies who considered this unfeasible because of the chance for a single point of failure. Interviewee 3 summarized it as so: “For me it was obvious that it should be decentralized, because to have such a centralized system not working means that the whole country shuts down, like a virus you know.”

Subsequently, they proposed a decentralized version of X-Road and designed organizational data exchange to be synchronous, which the Estonian government ultimately decided upon. In this first iteration, technical interoperability and availability of X-Road was ensured by the implementation of a closed Domain Name Security Extensions (DNSSEC) and an emphasis on “data being king”. The lead developer of X-Road, Interviewee 4, clarified why X-Road protects the data itself through encryption and not the transportation route: “We decided that we need to protect data not channels. We don’t need special channels; we can use some public channels (public internet). We say that data is king, and we protect this data by signing our data.”

Therefore, X-Road’s initial vision was based upon the design of the public internet, which is inherently decentralized and encrypted, along with some centralized organizational elements. However, the primary reason was due to lack of resources and infrastructure for private networks, not because of the internet’s primacy. At the time, the Estonian Information Systems Authority was considered the central “operator” of the X-Road ecosystem, onboarding other organizations and establishing governance. In its current technical form, X-Road uses an ecosystem of third-party trust services (timestamping, certificate authorities and Online Certificate Status Protocol services,) a central registry server of X-Road members

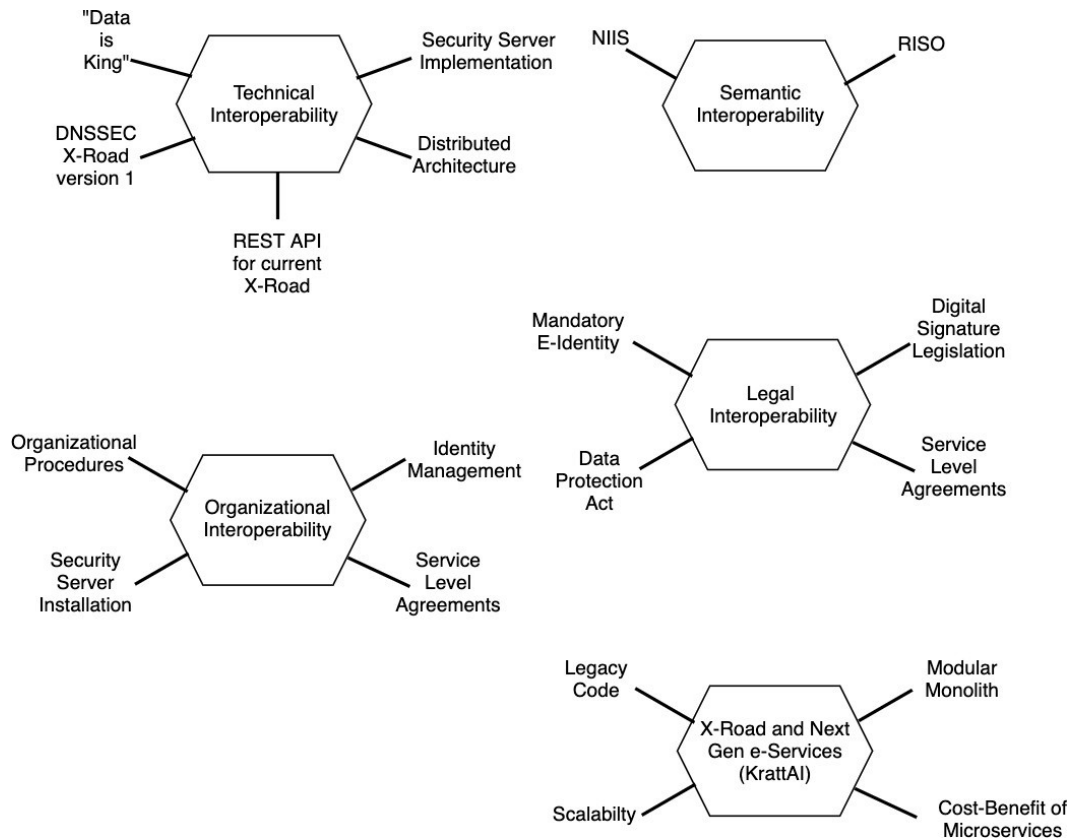


Figure 2: Thematic Analysis

and security servers that are configured to national, regional or local information systems [24]. Subsequently, each member can act as a service producer, consumer, or both [24]. Architecturally, X-Road supports both Simple Object Access Protocol (SOAP) and Representational State Transfer (REST) for exchanging synchronous requests of JavaScript Object Notation data (JSON), XML, text, RDF and other forms of data between X-Road entities based on service-level agreements between them.

4.2 Semantic and Organizational Interoperability

Semantically for X-Road, responsibility for creating a common corpus of understanding was put to the state information system department (RISO) of the Ministry of Economic Affairs. RISO produced a common repository which Interviewee 2 was leading at the beginning of the 2000s. This repository was an important knowledge base for all developers and architects, leading to synchronicity and semantic interoperability. Nowadays, NIIS provides this repository via GitHub [25] and online training sessions for Security Server installation and administration.

Historically and presently, the X-Road ecosystem relies upon organizational interoperability. All organizations in the X-Road ecosystem follow established procedures for Security Server installation. In the opinion of Interviewee 4, “legal contracts (service

level agreements) are like maybe 1/5 of the whole organizational complexity” in the X-Road ecosystem. The other components are related to organizational identity management: “X-Road is specifically a Secure Exchange Network, where you can be sure that the identity of an organization that is claimed is what this organization claims to be. This information system claims to be this information system. This is the identity you are managing in the center, so you must build yourself an understanding of what is the operator functionality: you are the Identity Manager of the systems. And if you are not fulfilling that position, your X-Road will never fly, you will just get another technical solution that means nothing because there is no organizational responsibility.”

Interestingly as Interviewee 4 points out, the number of interoperable organizations who use X-Road has actually been inflated. Before X-Road version 6 migration, the number of registered organizations in the ecosystem was at 1200 and using memory. Currently, there are now 600 registered organizations and thus almost half of the organizational membership “didn’t use or didn’t need X-Road”. Consequently, “the ecosystem was full of that kind of bloke and fat that wasn’t very useful.” Nonetheless, the migration to X-Road version 6 improved interoperability because it ensured that organizations who want to be an active part of the ecosystem are still participating.

4.3 Interoperability

Legal interoperability has been a cornerstone empowering X-Road's utility. Legislatively, the first version of X-Road was consulted in lockstep with Estonia's data protection authority. In parallel at the time, crucial legislation in the early 2000s made e-Identification obligatory for Estonian citizens and established digital signatures as equivalent to hand-written ones. These two legislative developments enabled X-Road's signing and logging scheme. Additionally, the Estonian Database Act was incorporated into the Public Information Act, enshrining X-Road as the main infrastructure of data exchange in the public sector through the State Register of Databases [26].

4.4 X-Road and Next Generation E-Services

The second objective of this paper is to answer the research question: How is the current X-Road ecosystem equipped for the future of next generation government services? As the CTO of NIIS, Interviewee 6 was asked specifically about potential microservice integration into X-Road. Asynchronous messaging between service producers and consumers is at the core of the KrattAI vision proposed by the Estonian government. Therefore, understanding the perspective of NIIS on this issue is important for developing future novel e-services in Estonia and abroad.

When asked if X-Road intends to use microservices in the future, Interviewee 6 pointed out that after X-Road's governance was officially transferred over to NIIS in June of 2018 they inherited its code base and responsibilities for maintenance and development. Interviewee 6 described this code as being a "modular monolith", because "when you build X-Road, you have to build the whole monolith. But then when we think about the Security Server, it consists of different modules that run as independent processes." The Security Server's modules mostly communicate through Akka based TCP protocol. Whereas in microservices, according to Interviewee 6, generally use HTTPS for interfacing. For X-Road, the vital question is how to support asynchronous messaging between service consumers and service producers in its current synchronous form. Microservices provide an architectural framework for achieving this aim, however there are other architectural patterns that can support asynchronous messaging that are not microservice driven.

Ultimately, the required changes to X-Road's architecture for implementing microservices requires too many resources and the cost-benefit of such a transition doesn't make business sense. As Interviewee 6 aptly describes, "It would be nice to say that we use microservices, we use blockchain and whatever is hot right now, but unfortunately we are a little bit boring. And that's why, at least not yet, we haven't gone into microservices."

Still, Interviewee 6 noted that microservices implementation would make a positive impact on X-Road's scalability. Bottlenecks can occur when there is a higher load pertaining to signing messages and logging. In order to scale, server capacity has to be increased, which means adding more resources to a single Security Server (scale up) or adding more Security Server instances (scale out). When scaling up, there is a limit where increased resource capacity to Security Servers doesn't meet scaling needs. As Interviewee 6 explains: "With microservices-based approach, especially when you run the whole system in containers, it might be enough

to scale out the container that is responsible for signing the messages instead of scaling out the whole Security Server and all the components. So, it would make it a lot more efficient."

5 DISCUSSION

From the interview data, it is apparent that the historical foundations of X-Road considered a decentralized interoperability model, where instead of expensive VPNs and high infrastructure costs, the public internet was used as the primary communication channel. This was due primarily to pragmatism and a lack of IT capital at the time. Starting with ten public sector databases and then interconnecting them through the public internet with specific logging and electronic signing functionality used for auditing and accountability. Vital to the e-signature format integrated into the X-Road was e-Identification and digital signature legislation.

Without these two pieces of legislation, the ability for X-Road to have legitimacy would be inhibited, at least in Estonia. This is because when Estonia made e-Identification mandatory it created an incentive mechanism to use the X-Road organizationally. It should be noted that the Estonian law on digital signatures has now been replaced by the EU's EIDAS legislation, which X-Road already abided by domestically. Semantically, X-Road's development has been quite straightforward. The establishment of RISO provided clear guidelines, standards and frameworks for developers to use in the early 2000s. This has now been updated through NIIS who provides the guidelines through their GitHub and governance practices.

An interesting facet of the interview data was the premise that technology is not the primary inhibitor of interoperability in the X-Road. Rather, it is organizational practices and internal policies that become major barriers. A good example of this was the organizational glut that occurred when migration from X-Road version 5 to version 6 (it's current iteration) was initiated. If over half of the existing ecosystem was a member by name only, then interoperability was reduced based upon organizations not "buying-in" to the concept of X-Road. A transition to where X-Road is streamlined organizationally can only help strengthen the interoperability environment that is currently implemented.

Lastly, the introduction of microservices to the X-Road is not because of a lack of understanding of but purely based on organizational objectives, as it is currently not on the NIIS Roadmap. It is important to understand the division between X-Road's internal architecture and messaging protocols and the architecture used by service producers and consumers in the ecosystem. X-Road producers and consumers can have microservice capability as long as they comply with X-Road's messaging protocols. In the end, NIIS is exploring how asynchronous messaging patterns can be implemented, regardless if it is microservices or not.

As Interviewee 4 elaborated, microservices do not have to be integrated directly into the X-Road, which would only complicate things as they are currently. Instead, a top layer microservice component could be implemented that masks the synchronicity of X-Road into an asynchronous one. This would allow the simplicity of X-Road to continue, while still providing architectural requirements needed for the virtual assistant e-services proposed by the Estonian government.

6 CONCLUSION AND FUTURE WORK

This research has mapped out some of the historical interoperability conditions that led to X-Road's implementation in Estonia, while also addressing its potential for supporting next generation digital government services. It is clear X-Road empowers interoperability and thus Estonian integrated e-services through the EIF paradigm. Technical interoperability is ensured through Security Servers and REST/SOAP APIs. Semantically, RISO was established in the early 2000s and NIIS's GitHub provides current specifications, documentation and source code for organizations. Interoperability among organizations in the X-Road ecosystem relies on identity management, use of Security Servers and a multitude of service level agreements between the X-Road member organizations. Lastly, legal interoperability for the X-Road in Estonia was provided by Estonian legislation on data protection, e-signatures and e-identity in the early 2000s.

Future possible developments involve creating a microservice solution for a cross-border virtual assistant e-service between Estonia and Finland, as there is a close cross-border relationship between both societies. This microservice solution might not be integrated directly into X-Road, but rather work as a top layer component for enabling the next generation of cross-border e-services. As X-Road is jointly funded by the Estonian, Finnish and Icelandic governments through NIIS, the above-mentioned future opportunities support cross-border data exchange between NIIS member countries and Latin America. A preliminary survey to cross-border e-service providers in the Finnish and Estonian public sectors could further solidify what use-case is appropriate for the above microservice layer.

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